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In [10]: import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

# Load data
df = pd.read_csv("/Users/tejabattina/Downloads/05042026/manufacturing_6G_data.csv")

# Features
features = [
    'Temperature_C', 'Vibration_Hz', 'Power_Consumption_kW',
    'Network_Latency_ms', 'Packet_Loss_%',
    'Quality_Control_Defect_Rate_%',
    'Production_Speed_units_per_hr',
    'Predictive_Maintenance_Score',
    'Error_Rate_%'
]

X = df[features]

# Standardization
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

# Convert to DataFrame
pca_df = pd.DataFrame(X_pca, columns=['PC1', 'PC2'])

# Add label
pca_df['Efficiency_Status'] = df['Efficiency_Status']

```

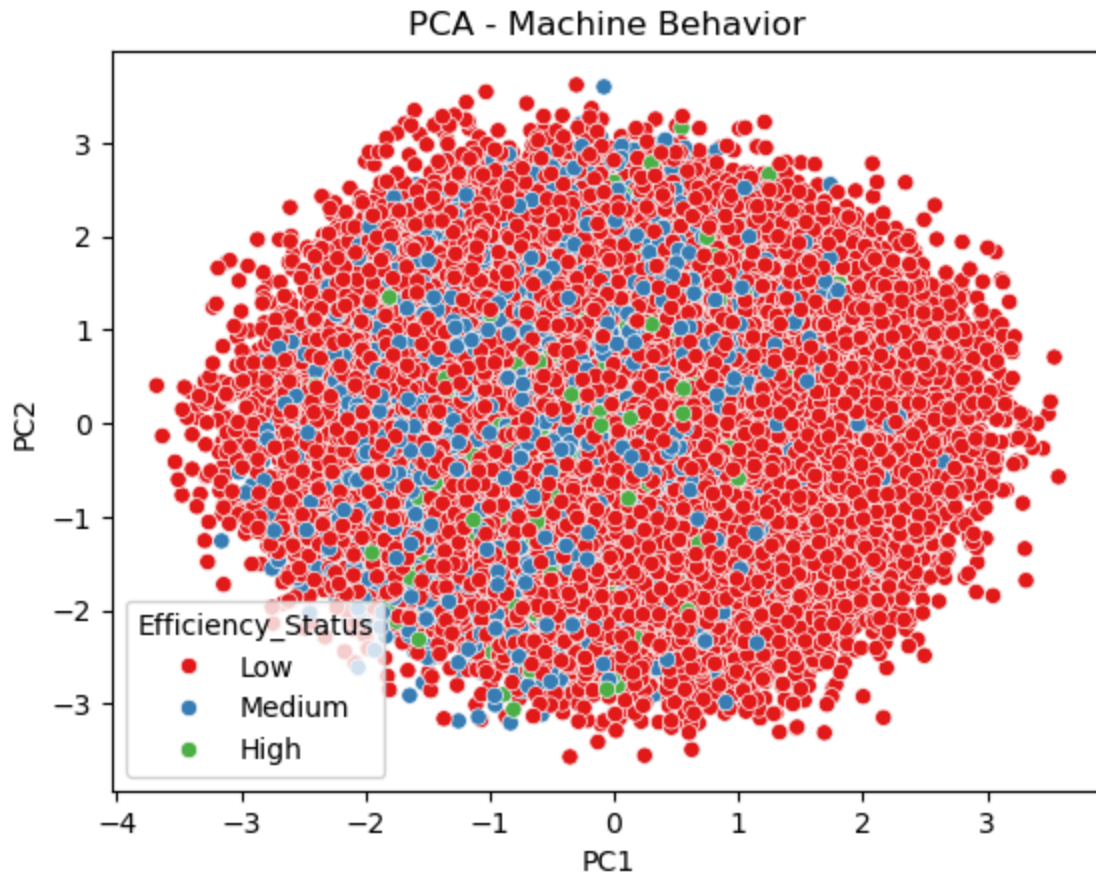
```

In [11]: import seaborn as sns

sns.scatterplot(
    x='PC1',
    y='PC2',
    hue='Efficiency_Status',
    data=pca_df,
    palette='Set1'
)

plt.title("PCA - Machine Behavior")
plt.show()

```



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In [12]: print("Explained variance ratio:", pca.explained_variance_ratio_)
print("Total variance captured:", sum(pca.explained_variance_ratio_))
```

Explained variance ratio: [0.11280493 0.11222922]
Total variance captured: 0.22503414664209212

```
In [13]: loadings = pd.DataFrame(
    pca.components_,
    columns=features,
    index=['PC1', 'PC2']
)

print(loadings)
```

	Temperature_C	Vibration_Hz	Power_Consumption_kW	Network_Latency_ms
PC1	-0.073682	0.388031	0.213921	0.283927
PC2	-0.438015	0.008739	-0.426994	0.539397

	Packet_Loss_%	Quality_Control_Defect_Rate_%
PC1	-0.376405	-0.153709
PC2	-0.415217	0.212593

	Production_Speed_units_per_hr	Predictive_Maintenance_Score	Error_Rate
PC1	0.460624	0.314921	0.4908
PC2	-0.176251	-0.271838	-0.1105

In []: